Machine Learning for Customer Churn Prediction

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1 Executive Summary

Unsupervised learning provides techiniques useful for different problems that complement the projects. The re-creation of existing data, the anomaly detection or generate new data similar to existing data. Can be help in the domain of Churn Prediction, in special for working with a dataset like the KKBox selected for this project.

2 Motivation

The contribution of this workshop to the project, is implement unsupervised learning for strategy parts in the Customer Churn Prediction system. There is a problem with the unbalanced dataset, then the experiment will consist in explore the AE and VAE for show how good reply existing customers and then implement for generate customer churns.
Purpose of this new customers generated with real customer churns, is generate more data and get a better performace, that the models of w1 got. The experiment will conclude if this whether or not this is feisible depends on the results.

3 AE and VAE

Autoencoder and variational autoencoder are unsupervised neural network arquitectures

3.1 Data and preprocessing

The autoencoders reuse the tabular export built in Workshop 1 (version 5), where the *members*, *transactions* and *user_logs* tables are condensed into a single record of 59 features per customer (56 numerical and 3 categorical). The February cohort (992,931 customers, 6.39% churn) is used for development and the March cohort (970,960 customers, 8.99% churn) as the held-out test set; 881,701 customers appear in both months.

The Workshop 1 split (stratified and hash-based on `msno`) is reused so that the same customers are used across workshops. The churn label is never fed to the networks; it is only used to select the training population. The AE and the VAE are trained on the 697,095 non-churn customers of the *fit* partition

and validated on the 116,183 non-churn customers of the *stop* partition, while the full March cohort, including churners, is used for testing.

A preprocessor is fitted on the training population only. Numerical features are imputed with the median, skewed features are transformed with a signed $\log(1+|x|)$, clipped to the 0.1% and 99.9% quantiles and standardized; one constant column was removed, leaving 55 numerical inputs. The categorical features (`city`, `registered_via`, `last_payment_method_id`) are mapped to vocabularies of 22, 6 and 34 levels and one-hot encoded inside the network, giving an input dimension of 117. An inverse transform maps reconstructions back to the original 59 columns. The AE and the VAE share the same fitted preprocessor.

3.2 AE Architecture

3.3 AE Train

3.4 VAE Architecture

3.5 VAE Train

4 Reconstruction quality

5 Anomaly detection

6 Latent space visualization

7 Generate synthetic data

8 Analysis

References

[1] Rizvi, A. (2018). *WSDM – KKBox's Churn Prediction Challenge*. Medium. Retrieved from `https://asad-99rizvi.medium.com/wsdm-kkboxs-churn-prediction-challenge-d7c9cfa21cbd`